Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-PICU-POL-9-Vers.1.0

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Title: Tracheostomy Care

Objectives:

This policy Aids at understanding the important aspects related to tracheostomycare in children which include theindications for tracheostomy inchildren, How to choose the size of tracheostomy tubes, post-operative care ,common complications related to tracheostomy placement and decannulationprocess.

Policy Statement:

This policy tobe used as a guide in PICU, PCSU, medical HD and cardiac HD areas.

Policy Scope:

This is to beused by nurses and doctors in intensive care areas.

Tracheostomy is defined as the formation of a stoma between thetrachea and the skin.

Indications for tracheostomy:

1. 1.Upper airwayobstruction which include: subglottic stenosis, tracheomalacia, trachealstenosis and craniofacial abnormalities such as Pierre Robin syndrome.
2. 2.Craniofacialand laryngeal tumors including cystic hygroma and hemangioma.
3. 3.Obstructivesleep apnea
4. 4.Laryngealtrauma
5. 5.Long termventilation
6. 6.Complications relatedto congenital heart disease such as diaphragmatic palsy
7. 7.Neurological/neuromuscular disorders leading to weakness

Tracheostomytubes should be selected based on the sizing of the trachea length and width. Thechoice of the size iscommonly based on the child age since other modalities likeradiological investigations and endoscopy are not routinely used in theseassessments.

This approach will fit most of children. Exception to that are children with shortstatue, short neck and airway abnormalities for which customized tracheostomytubessuch asBivona can provide an option.

**Reviewthe attached chart in thelast page of this document on sizing of the tracheostomytubes according to child age.

Thetube lenght should extend 2 cms beyond the stoma and should be above the carinaby 1-2 cms.

Achild who is less than 1 year, should be fitted with tracheostomy with neonatallength whereas, the child who is older than 1 year, should be fitted with atube of pediatric length.

A 15mm connector should be applied to outer diameter of the tracheostomy tube toaid in attaching the ventilator or the bag.

Tracheostomy tube composition:

Majority of thetubes used are the Plastic tubes (e.g. portex and Shiley) which are made ofpolyvinyl chloride. Bivona tubes are made of silicon and useful in childrenwith unusual anatomy.The metal tubes are not routinely used in clinicalpractice since these cannot be connected to the ventilators.

Cuffed tracheostomy tubes

Indications of cuffed tubes include:

· * Ventilationwith high pressures

· * Chronicaspiration

· * Nocturanlventilation where the cuff can be inflated at night and deflated when child isawake

Speaking valves:

The use of speaking valves helps children to phonate. Thesevalvescontain bias – closed diaphragm which opens during inspiration and closes atend of inspiration.

Contraindications to use of speaking valves:

- - Very young orvery ill children

- - Children atrisk of aspiration

Post-operative care:

After the child returns to intensive care, obtain CXR to review theposition of the tracheostomy tube (tip should lie above the carina) and toexclude pneumohorax.

Ensure that humidified air is used to reduce chances of thicksecretions and ensure tube patency.

The following equipments should be ready at the bed side:

· * Emergency Kitwhich includes:

-Anadditional tracheostomy tube of the same size and same brand

-Tracheostomytube one size smaller

-Tracheostomytapes

-Waterbased lubricants

-humidification

-suctioncatheters

The first tube change is performed after 7 days **. The subsequenttube changes are done according to the unit practices and the patient needs.

** There may be differences to this practice.

Post-operativecomplications:

Complications related to tracheostomy range between 0.5%-3%. Themost serious complications include blockage of tracheostomy tube and accidentaldecannulaion of the tube.

Immediate post-operative complications:

Accidental decannulation of the tube:

This is associated with increase morbidity and mortality. Thiscomplication can be prevented by appropriate tube selection, appropriatesedation, careful nursing and handling of the tube and presence of securetapes.

All staff taking care of a tracheostomised child should be familiarwith accidental decannulation plan.

** Pleaserefer to the unit algorithm of how to handle accidentaldislodgement of the tube.

Blockage of the tracheostomy tube

This is seen commonly in smaller children. Frequent tube care canavoid this complication.

Bleeding:

This is one of the serious complications. This can be caused by theirritation of the tube as it comes in contact with tracheal wall. In seriouscases, this may reach innominate artery and leads to life threatening bleeding.Bleeding from tracheostomy tube mandates endoscopic evaluation

Tracheo-esophgeal fistula:

This is may be caused by contact irritation of the posteriortracheal wall which can lead to life threatening contamination of the airwaypassages.

Late complications

Late complications are commoner than early complications and arereported in 60% of children.

Suprastomal granulation:

This is common. It is seen in the superior margin of thetracheostome at the tracheal lumen 2nd to irritation of the tube tothe mucosa. Large and obstructing granuloma needs treatment prior todecannulation.

Suprastomal collapse:

This is seen in chronic patients where chronic irritation by thetube leads to inflammation of the cartilage rings and destruction. The collapseis seen in the anterior tracheal wall superior to the tracheostome.

Significant collapse needs treatment before decannulation.

Subglottic / tracheal stenosis:

Both of these are described in chronic patients. Subglotticstenosis is seen as result of intraoperative damage of the cricoids or 2ndto high tracheostomy tube.

Decannulation process:

Decannulation is considered once the cause for tracheostomy hasresolved or treated. This process to be done in consultation with ENT team.

Laryngoscopy and endoscopy need to be performed to exclude presenceof granulation tissue or suprastomal collapse which need treatment.

If the above are treated, the child can be decannulated with thefollowing steps:

· * Down size thetracheostomy tube to the smallest size

· * Occludetracheostomy for 12 hours.

· * If successful ,occlude the tracheostomy overnight

· * If successful ,the child can be decannulated and observed

			Preterm-1 month		1-6 months	6-18 months	18 mths - 3 yrs	3-6 years	6-9 years	9-12 years	12-14 years
	Trachea (Transverse Diameter mm)		5		5-6	6-7	7-8	8-9	9-10	10-13	13
PLASTIC	Great Ormond Street	ID (mm)		3.0	3.5	4.0	4.5	5.0	5.5	6.0	7.0
		OD (mm)		4.5	5.0	6.0	6.7	7.5	8.0	8.7	10.7
	Shiley	Size		3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5
		ID (mm)		3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5
		OD (mm)		4.5	5.2	5.9	6.5	7.1	7.7	8.3	9.0
		Length (mm) Neonatal		30	32	34	36				
	*Cuffed Tube Available	Paediatric		39	40	41*	42*	44*	46*		
		Long Paediatric						50*	52*	54*	56*
	Portex (Blue Line)	ID (mm)		3.0	3.5	4.0	4.5	5.0	5.0	6.0	7.0
		OD (mm)		4.2	4.9	5.5	6.2	6.9	6.9	8.3	9.7
	Portex (555)	Size		2.5	3.0	3.5	4.0	4.5	5.0	5.5	
		ID (mm)		2.5	3.0	3.5	4.0	4.5	5.0	5.5	
		OD (mm)		4.5	5.2	5.8	6.5	7.1	7.7	8.3	
		Length Neonatal		30	32	34	36				
		Paediatric		30	36	40	44	48	50	52	
	Bivona	Size	2.5	3.0	3.5	4.0	4.5	5.0	5.5		
		ID (mm)	2.5	3.0	3.5	4.0	4.5	5.0	5.5		
		OD (mm)	4.0	4.7	5.3	6.0	6.7	7.3	8.0		
	All sizes available with Fome Cuff, Aire Cuff & TTS Cuff	Length Neonatal	30	32	34	36					
		Paediatric	38	39	40	41	42	44	45		
	Bivona Hyperflex	ID (mm)	2.5	3.0	3.5	4.0	4.5	5.0	5.5		
		Usable Length (mm)	55	60	65	70	75	80	85		
	Bivona Flextend	ID (mm)	2.5	3.0	3.5	4.0	4.5	5.0	5.5		
		Shaft Length (mm)	38	39	40	41	42	44	45		
		Flextend Length (mm)	10	10	15	15	17.5	20	20		
SILVER	Alder Hey	FG		12-14	16	18	20	22	24		
	Negus	FG			16	18	20	22	24	26	28
	Chevalier Jackson	FG		14	16	18	20	22	24	26	28
	Sheffield	FG		12-14	16	18	20	22	24	26	
		ID (mm)		2.9-3.6	4.2	4.9	6.0	6.3	7.0	7.6	
	Cricoid (AP Diameter)	ID (mm)		3.6-4.8	4.8-5.8	5.8-6.5	6.5-7.4	7.4-8.2	8.2-9.0	9.0-10.7	10.7
	Bronchoscope (Storz)	Size		2.5	3.0	3.5	4.0	4.5	5.0	6.0	6.0
		ID (mm)		3.5	4.3	5.0	6.0	6.6	7.1	7.5	7.5
		OD (mm)		4.2	5.0	5.7	6.7	7.3	7.8	8.2	8.2
	Endotracheal Tube (Portex)	ID (mm)	2.5	3.0	3.5	4.0	4.5	5.0	6.0	7.0	8.0
		OD (mm)	3.4	4.2	4.8	5.4	6.2	6.8	8.2	9.6	10.8

Reproduced with Permisson from Wyatt et al.[6]

References:

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
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